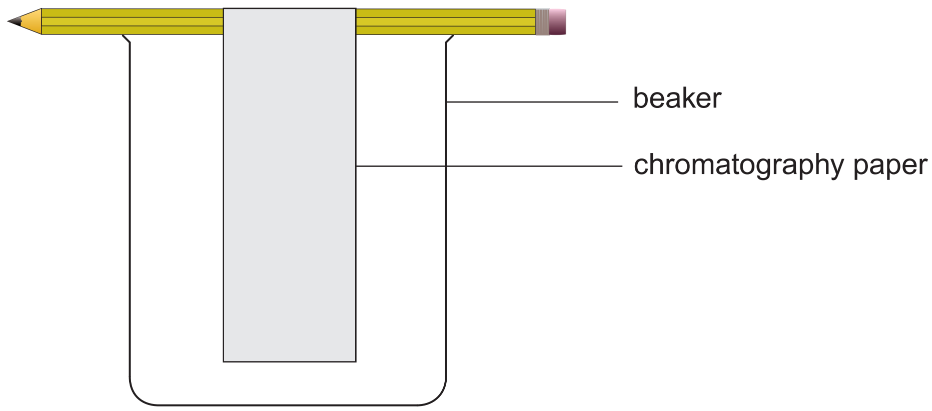
Examiner only

2. (a) Amanda wanted to determine what coloured dyes were present in a sample of orange ink.

The diagram shows a piece of chromatography paper, supported by a pencil, placed in a beaker at the start of her experiment.

Complete the diagram by showing

- the position of the ink sample at the start
- the water level in the beaker [2]



(b) The table shows the R_f values for some coloured dyes that are found in inks.

Dye colour	R_f value
blue	0.40
yellow	0.25
red	0.70
green	0.15

(i) Explain why coloured dyes have different R_f values. [2]

....

....

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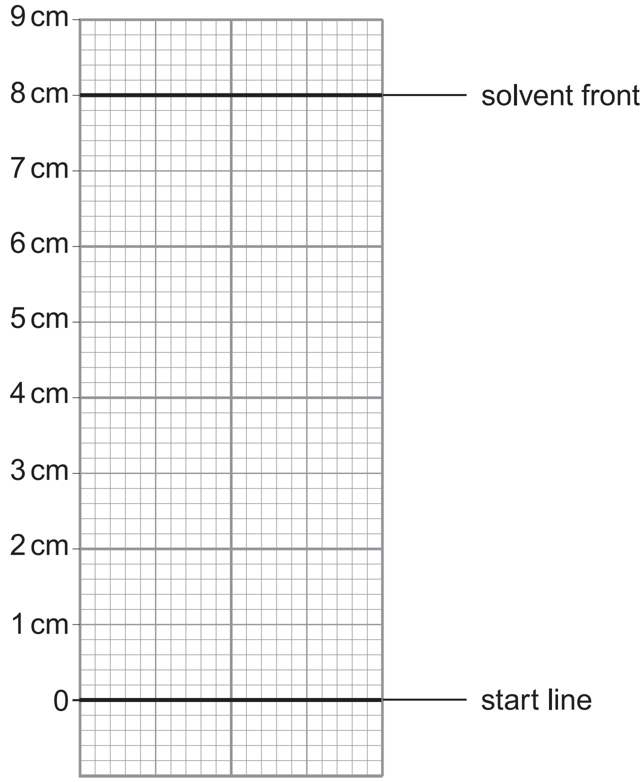
Examiner only

(ii) Orange ink separates into red and yellow dyes.

On the chromatogram, draw the positions of the spots you would expect to see after a sample of orange ink has been analysed by chromatography. [2]

Use the formula

$$\text{distance travelled by dye} = R_f \text{ value} \times \text{distance travelled by solvent}$$



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